Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
  Black: Molex 2147601225
  Red: Molex 2147602225

Reverse Polarity Protection

Q9
CSD19534Q5A

J10
POWER
2 V_IN
1 V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C54
10uF-50V-10%-X7R-1210

C52
0.1uF-50V-10%-0603

U7
LM74700QDBVRQ1
A G C
VCAP
EN
GND

C45
10uF-50V-10%-X7R-1210

R28 5.1k-1%-0603
R27 5.1k-1%-0603
R26 5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
15V 3A Buck-Boost Power Supply
+V_OUT
-V_OUT

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C11
0.1uF-50V-10%-0603

U1
LM74700QDBVRQ1
A G C
VCAP
EN
GND

C12
10uF-50V-10%-X7R-1210

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
  Black: Molex 2147601225
  Red: Molex 2147602225

J4
V_OUT 2
V_OUT_RTN 1
15V
Molex
0436500200

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
4
3
2
1
Molex
5040500491

File: canable2.kicad_sch
3/5
CANable2.0
USB_DP
USB_DM
3V3_USB
GND_CAN
CAN_HI
CAN_LO

J9
2
1
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal

J2
Terminal
Keystone Electronics
7774

J3
NX CONTROL & UART
1 3V3_BTN
2
3
4 NX_PC_LED-
5 NX_PC_LED+
6 UART2_RXD
7 UART2_TXD
8 AC OK
9 Auto Power On Disable
10
11 SYS_RESET
12
13 REC
14 GND
15 SLEEP/WAKE*
Molex
PH1RB-15-UA
GND1

3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
NX_PC_LED+
NX_PC_LED-

D3
PC

SW2
RECOVERY

SW1
RESET

GND1

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

File: usb-serial.kicad_sch
5/5
USB Serial
3V3
RXD
TXD
UART2_TXD
UART2_RXD

H1 MountingHole
H5 MountingHole
H6 MountingHole
H2 MountingHole
H7 MountingHole
H8 MountingHole
H9 MountingHole

H10 1.152mm ToolingHole
H11 1.152mm ToolingHole
H12 1.152mm ToolingHole
H13 1.152mm ToolingHole

Light Pipe
LP1 Bivar Bivar-SLP3-150-100-F
LP2 Bivar Bivar-SLP3-150-100-F
LP3 Bivar Bivar-SLP3-150-100-F
LP4 Bivar Bivar-SLP3-150-100-F

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1868 SPACE COOKIES

Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-10	Rev: 7
KiCad E.D.A. 10.0.7		Id: 1/5

+V_IN
−V_IN
+V_OUT
−V_OUT
GND
Vcc

C59 47uF 50V
C58 47uF 50V
C44 10uF−50V−10%−X7R−1210
C9 10uF−50V−10%−X7R−1210
C6 10uF−50V−10%−X7R−1210
C8 10uF−50V−10%−X7R−1210
C10 10uF−50V−10%−X7R−1210
C43 10uF−50V−10%−X7R−1210
C49 10uF−50V−10%−X7R−1210
C50 10uF−50V−10%−X7R−1210
C7 10uF−50V−10%−X7R−1210

HDRV1 Q8 CSD19534Q5A
LDRV1 Q6 CSD19534Q5A
Q3 CSD17578Q5A HDRV2
Q4 CSD17578Q5A LDRV2
SW1 L1 SW2
2.2uH−19A−5.1mohm−20%

RCS
R14 10mOhm−3W−1%−2512
CSG

C15 10uF−50V−10%−X7R−1210
C19 10uF−50V−10%−X7R−1210
C18 10uF−50V−10%−X7R−1210
C14 10uF−50V−10%−X7R−1210
C24 10uF−50V−10%−X7R−1210
C23 10uF−50V−10%−X7R−1210
C2 10uF−50V−10%−X7R−1210
C4 10uF−50V−10%−X7R−1210
C5 10uF−50V−10%−X7R−1210
C3 10uF−50V−10%−X7R−1210

D8 BAT54KFILM
R21 249K−1%−0603
R19 93.1K−1%−0603
R17 2K−1%−0603
R20 10R0−1%−0603
C39 0.1uF−50V−10%−0603

Texas Instruments
LM5176PWPR
U4
VIN
VISNS
EN/UVLO
SS
COMP
MODE
RT
DITH
SLOPE
CS
CSG
VCC
BIAS
VOSNS
HDRV1
BOOT1
SW1
LDRV1
FB
PGOOD
HDRV2
BOOT2
SW2
LDRV2
ISNS(+)
ISNS(−)
AGND
PGND
EP

C37 0.1uF−50V−10%−0603
R22 0R−1%−0603 HDRV1
R18 0R−1%−0603 LDRV1
R10 0R−1%−0603 HDRV2
R13 0R−1%−0603 LDRV2
D7 BAT54KFILM
C36 1uF−50V−10%−0603
C40 0.1uF−50V−10%−0603
D6 BAT54KFILM
C29 0.1uF−50V−10%−0603

R3 357K−1%−0603
R5 20K−1%−0603

C33 15nF−10%−50V−0603
C26 5.6nF−10%−50V−0603
R6 5.9K−1%−0603
C28 5.6nF−10%−50V−0603
C31 270pF−10%−50V−0603
R16 205K−1%−0603
R15 13K−1%−0603
C35 150pF−50V−10%−0603
C27 47pF−50V−10%−0603
R8 100R−1%−0603
R7 100R−1%−0603

VIN(MAX)	24V
VIN(MIN)	5V
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V−3A−Power Supply Analysis

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Sheet: /15V 3A Buck−Boost Power Supply/
File: buck−boost−power−supply.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026−10−10	**Rev: 7**
KiCad E.D.A. 10.0.7		Id: 2/5

5V-250mA-Power Supply Analysis

Reverse Polarity Protection

Q2
SI2301CDS-T1-GE3

3V3_USB

3V3

L2
2.2uH-41mOhm-20%

U2
TPS613222ADBV
SW
VOUT
GND

5V_CAN

C13
22uF-10%-16V-0805

C16
22uF-10%-16V-0805

C17
22uF-10%-16V-0805

GND_CAN

Q7
SI2301CDS-T1-GE3

3V3

3V3_CAN

R9
1M-1%-0603

D10
BAV199LT1G

R12
93.1k-1%-0603

Q5
SS8050

C30
22uF-10%-16V-0805

GND_CAN

3V3_CAN delays bringing CAN RX (BOOT0) high until NPN turns on. ~10sec.

3V3

LQFP

R4
13K-1%-0603

C25
0.1uF-50V-10%-0603

GND_CAN

WORD_LED

VDDA
VDD
VDD
VDD
VBAT
VREF+
NRST
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PG10
PF0
PF1
PC13
PC14
PC15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB10
PB11
PB12
PB13
PB14
PB15
USB_DM
USB_DP
SYS_JTMS-SWDIO
SYS_JTCK-SWCLK
FDCAN1_RX
FDCAN1_TX
VSSA
VSS

USB_DM
USB_DP
SWDIO
SWCLK
STAT_LED

CAN_RX
CAN_TX
BOOT0

R11
13K-1%-0603

U3
STM32G431CBTx

GND_CAN

5V_CAN

C53
0.1uF-50V-10%-0603

GND_CAN

U8
TJA1051T/3/1J
TXD
RXD
VIO
S
VCC
GND
CANH
CANL

CAN_TX
CAN_RX
3V3_CAN

C48
0.1uF-50V-10%-0603

GND_CAN

CAN_H
CAN_L

CAN_H
CAN_L

D9
PESD1CAN,215

GND_CAN

3V3

D5
WORD

D4
STAT

R2
1K-1%-0603

R1
1K-1%-0603

WORD_LED
STAT_LED

GND_CAN
SWDIO
SWCLK
BOOT0
3V3

J8
SWD
PH1RB-05-UA
Adam Tech

3V3

C22
0.1uF-50V-10%-0603

C20
0.1uF-50V-10%-0603

C21
0.1uF-50V-10%-0603

C34
0.1uF-50V-10%-0603

C32
0.1uF-50V-10%-0603

GND_CAN

CANable 2.0 Schematic

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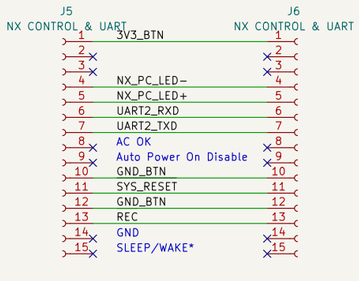
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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-10	**Rev: 7**
KiCad E.D.A. 10.0.7		Id: 3/5



H3
M3.7 MountingHole

H4
M3.7 MountingHole

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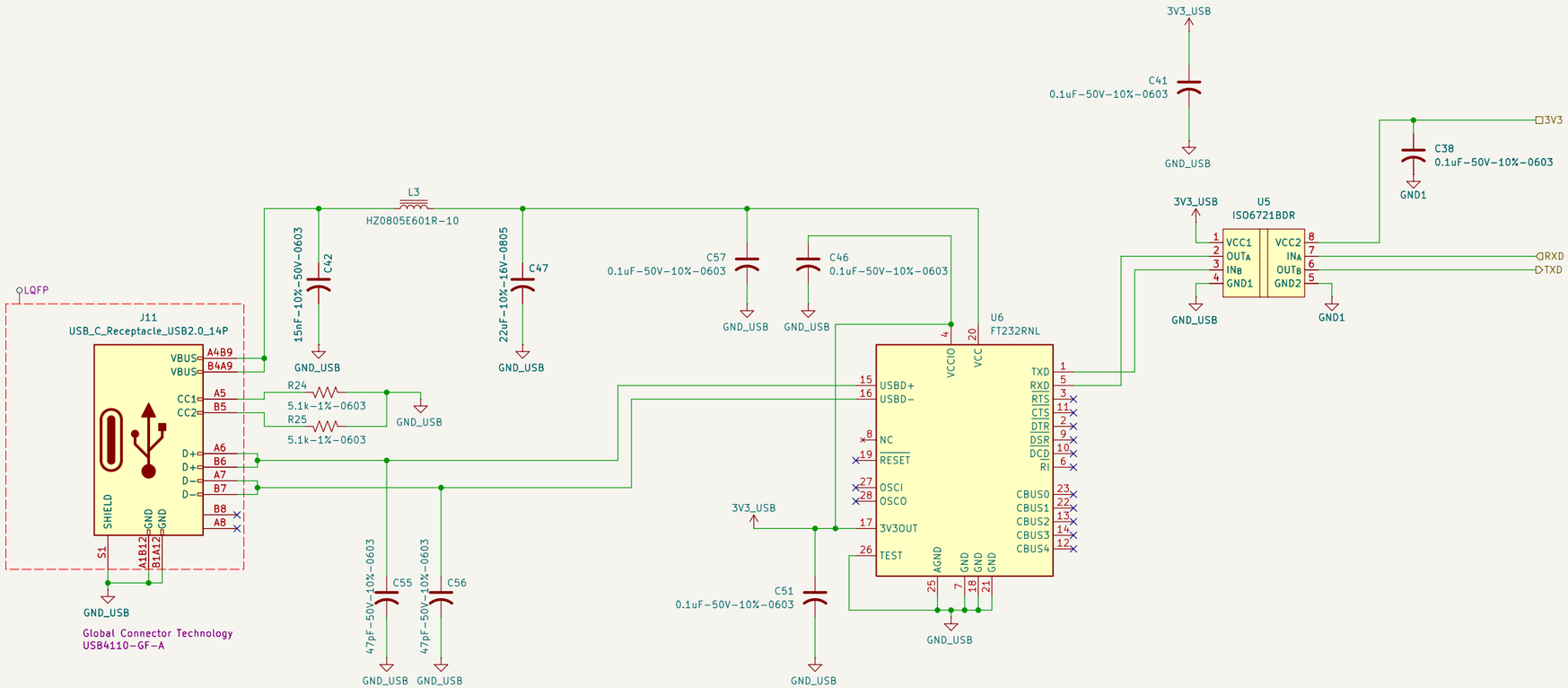
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Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-10	**Rev: 7**
KiCad E.D.A. 10.0.7		Id: 4/5



https://hackaday.com/2023/08/07/all-about-usb-c-example-circuits/

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1868 SPACE COOKIES

Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-10	**Rev: 7**
KiCad E.D.A. 10.0.7		Id: 5/5